



Fig. 2: Celestia displaying orbital path of planets

reality, at any speed, in any direction and at any time in history. It displays and interacts with objects ranging in scale from small spacecrafts to entire galaxies in three dimensions using OpenGL—from perspectives which would not be possible from a classic planetarium or other ground-based display. NASA and ESA have used Celestia in their educational and outreach programs as well as for interfacing to trajectory analysis software.

Celestia displays Hipparcos Catalogue of 118,322 stars and uses the very accurate VSOP87 theory of planetary orbits. This makes it possible for it to provide a solar and lunar eclipse finder and display the orbital paths of planets (including extrasolar planets), dwarf planets, moons, asteroids, comets, artificial satellites and spacecrafts. The users can vary the number of stars that are visible on the screen and have them drawn in different styles.

Celestia users can travel/fly through the Celestia universe using simple keyboard controls. Viewpoints can be set to look forward, backward or at any angle to direction of travel. Controls allow users to orbit stars, planets, moons and other space objects; track space objects such as spacecrafts, asteroids and comets as

they fly by; or travel to and/or fly through nebulae and irregular, elliptical and spiral galaxies (over 10,000 galaxies included).

The time simulated by Celestia can be set at any point in the future or past, although planetary orbits are accurate only within a few thousand years of the present day, and date arithmetic overflows at the year 5874774. The names and positions of multitudes of objects in space, as well as the names and locations of cities, craters, observatories, valleys, landing sites, continents, mountains, seas and other surface features can be displayed.

Celestia displays such features as detailed atmospheres on planets and moons, planet shine on orbiting satellites, sunsets and sunrises, moving clouds, planetary rings, eclipse and ring shadows, constellation lines, borders and illustrations, night-side lights of cities, detailed surface textures, specular reflections off water and ice, nebula gases and star flares.

Information about the objects that Celestia draws can also be displayed. This includes the radius, distance, length of the sidereal day and average blackbody temperature of the planets, as well as the distance, luminosity relative to the sun, spectral class, surface temperature and radius of stars.

The users can change Celestia's field-of-view from as wide as 120 degrees to a highly magnifying 3.4 seconds of arc, while dividing the window into multiple panes, in order to observe several objects at the same time. Graphic screenshots and movies can be captured in classic or HD resolutions (up to 1920x1080) on Windows and Linux platforms.

Celestia can be extended with new objects. There are hundreds of third-party, user-created add-ons (both fictional and realistic) available for installation. The extension mechanism uses Lua as its built-in scripting language.

TurboCASH

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